

Alec Portelli

Active U.S. Secret Clearance

alecportelli@cloud.com

alecportelli.com

617-455-1077

github.com/alecdportelli

Work Experience

The MITRE Corporation

Autonomy Software Engineer II • *April 2023 - Present*

Autonomy Software Engineer I • *July 2021 - April 2023*

- Leading the development of an autonomous underwater vehicle for the Navy. Utilizing NVIDIA Omniverse to train the vehicle to complete real life missions with AI and digital twins in Python and C++.
- Currently building a testbed that enables customers to simulate underwater vehicles in the MITRE maritime test facility. Writing code to help sponsors train machine learning models for AUVs using MITRE's high performance compute capabilities to increase company revenue.
- One of three engineers who built and delivered a Navy sponsored autonomy challenge for graduate students at the University of Washington. Created a 6 degree of freedom submarine physics model, high fidelity SONAR simulations, and helped write the AI control algorithm for advanced navigation.
- Conducted over ten live technical demonstrations to NVIDIA, the Navy, Microsoft, and MITRE executives which secured over a year of additional funding. Presented project findings at I/ITSEC 2023 to NVIDIA and Navy engineers. Led weekly meetings with the Office of Naval Research to ensure client needs were met while also staying with engineering constraints.

Selected Masters Program Projects

UAV Systems And Controls

Wrote a 6DoF simulation model of a quadcopter using Simulink and MATLAB. Built a model quadcopter from scratch using an Arduino. Designed and implemented a control system and wrote navigation algorithms.

Intro To Reinforcement Learning

Created a model UAV quadcopter in Unity and trained it to navigate an unknown environment. Used trained model to simulate package delivery systems amongst multiple UAVs. Built a neural network from scratch.

Engineering Skills

Areas Of Interest

Robot simulation, machine learning, 3D modeling, simulation, control theory, SLAM, digital twins

Programming Languages

C#, Python, MATLAB, Javascript, C++, Rust, Go

Tools

NVIDIA Omniverse, MATLAB, Simulink, ROS 2, PyTorch, Unity, Git, Git CI/CD, Docker, AWS, Arduino

Education

Johns Hopkins University - M.S. Robotics Engineering • *Expected May 2025*

Coursework includes: Algorithms for Data Science, Intro To Reinforcement Learning, UAV Systems and Controls, Kinematics and Dynamics For Robots.

Tufts University - B.S. Engineering • *Graduated May 2021*

Double major in human computer interaction and biomedical engineering. Graduated *cum laude*.